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arjunswaj

Arjun Shekar Bharadwaj

Software engineer with 10+ years of experience building high-scale distributed systems and cloud-native platforms in Java. Skilled in API design, design patterns and container orchestration.

Experience

2024–Present **Senior Software Engineer**, *Microsoft Azure*, Mountain View

Building secure, cloud-native compute platforms for Azure Resource Builder, processing billions of resource events for monitoring, compliance and automation.

- Azure Resource Builder Compute Platform
  - Separated control and data planes to automate workspace provisioning, resource deployment and compute lifecycle management
  - Integrated Apache Airflow for workflow-driven processing with explicit retries, error handling and partner-defined DAGs
  - Introduced containerized runtimes on Kubernetes, enabling partners to run Flink, Spark and RisingWave without language lock-in
  - Built the proof of concept, secured funding and led four engineers to deliver the platform with self-service onboarding
- Developer Experience & AI Tooling
  - Created container solution for dynamic NuGet package updates with zero-downtime deployments
  - Championed MCP servers for agentic workflows and custom prompts to speed up team development cycles
  - Deployed containerized services on AKS using Helm charts, with multi-stage CI/CD pipelines for build, test and release

2015–2024 **Software Development Engineer**, *Amazon*, Sunnyvale/Bengaluru

Built large-scale distributed services in Java (~80%) and Python (~20%) using Spring Boot and OOP design patterns. Delivered low-latency REST APIs, payment systems and multi-device platforms.

- AI-Powered Targeting & Action APIs – Alexa+
  - Built production AI agent system with MCP-like architecture. Enabled LLM-based device targeting for Music, News and Video domains
  - Created prompt engineering pipelines with In-Context Learning. Achieved **90%+ accuracy** through test harness evaluation
- Cost & Latency Optimization – Music Recommendations
  - Rebuilt rule engine using GoF patterns (Strategy, Composition). Cut database calls by 50% and **lowered costs by 50%** with optimized query patterns and improved tail latency
  - Shipped Java APIs handling millions of daily requests with tuned concurrency, connection pooling and retry-with-jitter to meet SLOs at peak load
- Multi-Device Platforms & Leadership
  - Served as tech lead for Vega OS working group with 20+ members (Principal Engineers, Senior Engineers, Product Managers). Drove API design alignment across device and cloud stacks
  - Ran Q4 readiness with load tests, chaos tests and capacity planning. Mentored engineers on GoF design patterns, JUnit testing and code quality
- Invoice Generation – Global Seller Platform
  - Built async invoice workflow generating millions of invoices and transfer it over a secure SFTP channel to Deutsche Bank for audits
  - Implemented connection pooling and Command pattern for fault-tolerant bulk transfers with 99%+ success rate

Education

2013–2015 **Master of Technology (M.Tech)**, *IIIT-B*, India, *CGPA: 3.70/4.00*
Major: Information Technology

2007–2011 **Bachelor of Engineering (B.E)**, *RVCE*, India, *CGPA: 9.00/10.00*
Major: Computer Science and Engineering

Skills

Languages	Java, Python, JavaScript, C#, Haskell
Frameworks & Testing	Spring Framework, Spring Boot, JUnit, Mockito, Proto + gRPC, Angular
Cloud Platforms	AWS (SQS, SNS, Lambda, DynamoDB, Athena), Azure (Event Hubs, Storage Accounts, Container Apps)
Databases	DynamoDB, Redis, PostgreSQL, MySQL
Container & Orchestration	Kubernetes, Docker, Helm, Azure Container Registry, CI/CD Pipelines
Data & Streaming	Apache Airflow, Apache Spark, Apache Flink, Kafka, AWS Data Pipeline
AI & ML	LLM Integration, Prompt Engineering, Agentic Workflows, MCP Servers, RAG Systems